UNIT – IV : SHALLOW FOUNDATIONS

1. Foundation – Meaning & Functions

- **Foundation** = the lowest part of a structure which transmits the load of the superstructure safely to the soil / rock below.
- Two basic requirements of every foundation:
  - Safety against shear failure of soil (bearing capacity, F.S. ≥ 2.5 to 3)
  - Serviceability – total & differential settlement within permissible limits

2. Classification of Foundations

- **Shallow foundation** : $D_f \leq B$ (Terzaghi); load is carried mainly by base resistance. Eg: spread, strip, combined footing, raft.
- **Deep foundation** : $D_f > B$ ($\geq$ 3 to 4 B); load carried by skin friction + end bearing. Eg: piles, piers, wells (Unit V).
- Here D_f = depth of foundation, B = least lateral dimension of base, L = length.

3. Types of Shallow Foundations

(a) Isolated (Spread / Pad) Footing

- Under a single column; plan square, rectangular or circular. Most common and cheapest type.
- Types of section: plain concrete (block), stepped, sloped (tapered) and pedestal footing; R.C.C. for heavy loads.
- Used when soil is of good bearing capacity (S.B.C $\geq$ 150 kN/m^2) and columns are far apart so that adjacent footings do not overlap.

(b) Strip / Wall Footing

- Continuous footing under a load-bearing wall ($L \gg B$, plane-strain condition). Also used for a row of closely spaced columns.
- Brick/stone masonry walls: stepped (offset) footing with offset $\leq$ ½ of the step height; spread $\approx$ 2 × wall thickness.

(c) Combined Footing

- Supports **two or more columns**. Provided when (i) columns are so close that isolated footings overlap, (ii) a column is near the property line and the footing cannot project beyond it, (iii) bearing capacity is low.

- Rectangular combined footing : when loads of both columns are nearly equal (or the column with larger load has more space)
- Trapezoidal combined footing : when loads are unequal; centroid of base area is made to coincide with the resultant of column loads so pressure is uniform.

(d) Strap (Cantilever) Footing

- Two isolated footings joined by a **strap beam**; strap carries the moment due to eccentric load of the exterior column so that the footings behave independently. Economical when distance between columns is large (a combined footing would be heavy).

(e) Raft (Mat) Foundation

- A large continuous slab covering the whole area under the structure and supporting all columns / walls.
- Used when : (i) soil is weak, S.B.C. low – isolated footings would cover > 50 % of the plan area; (ii) differential settlement must be reduced; (iii) compressible pockets / erratic soil; (iv) basement & water-proofing needed; (v) heavy loads.
- Types: flat plate raft, raft with beams (slab-and-beam), raft with pedestals, cellular raft, piled raft.
- **Floating (compensated) raft:** weight of soil excavated ≈ weight of the structure, so net pressure on soil is almost zero → no settlement problem. Net pressure $q_n = q - \gamma D_f$

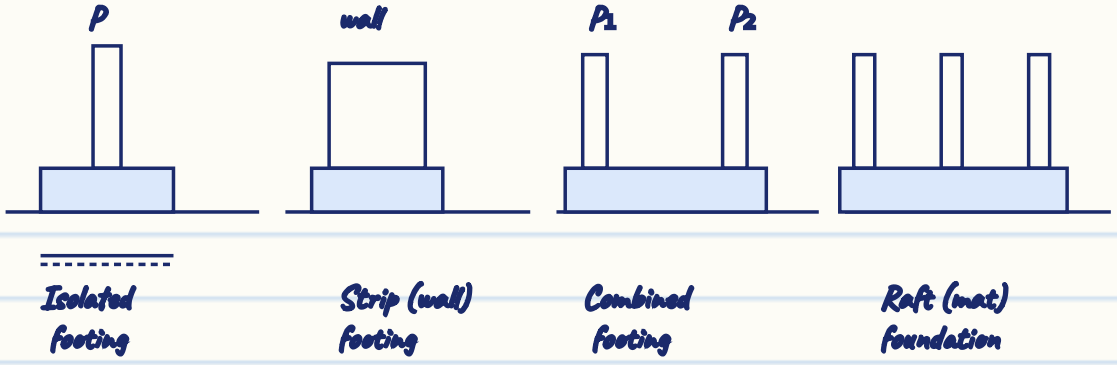


Fig. 1 – Common shallow foundations (side elevations)

5. Choice of Type of Foundation

Selection depends on a combination of factors – decided after soil investigation:

1. **Soil conditions** – type, S.B.C., compressibility, water table, presence of expansive / collapsible / filled soil.

2. **Magnitude & nature of loads** – dead, live, wind, seismic, eccentric / moment loads, dynamic loads.
3. **Type of superstructure** – flexible / rigid frame, sensitivity to differential settlement (framed building vs. tank).
4. **Depth of water table** & ground-water chemistry (sulphates, chlorides).
5. **Adjacent structures** – undermining, interference of stress bulbs, property line limits.
6. **Economy, construction time, available equipment**, local practice.

6. Depth & Location of Foundation

Depth D_f is governed by:

- **Bearing stratum** – founded below top soil / organic matter / filled ground, on a firm stratum. IS 1904: minimum depth 0.5 m below ground level.
- **Frost action** – below the frost line (in cold regions 0.6–1.5 m) to avoid heave.
- **Seasonal volume change** – in expansive (black cotton) soils go below the active zone ($\approx$ 1.5 – 3.5 m), or use under-reamed piles / sand cushion.
- **Scour** (bridges) – foundation at $\geq$ 1.2–2 m below maximum scour depth.
- **Water table** – above W.T. preferably, to avoid dewatering and buoyancy.
- **Adjacent foundations** – new footing should not be deeper than neighbour's unless it is at a distance; line of 2H : 1V (or 45°) from the old footing edge must be respected.
- **Underground facilities**, basements, lift pits, machine foundations.

Rankine's minimum depth (for granular soils)

$$D_f = (p / \gamma) \cdot [(1 - \sin\varphi) / (1 + \sin\varphi)]^2$$

- p = intensity of gross loading (kN/m^2), γ = unit weight of soil, φ = angle of shearing resistance. Gives the depth below which soil cannot be squeezed out laterally.

Location (plan): centroid of base must coincide with line of action of resultant load ($e \leq B/6$); keep clear of existing structures & slopes.

8. Bearing Capacity – Definitions

- **Ultimate bearing capacity (q_u)** : minimum gross pressure intensity at the base of the foundation at which the soil fails in shear.

- **Net ultimate bearing capacity (q_{nu})** : net increase in pressure over overburden : $q_{nu} = q_u - \gamma D_f$
- **Net safe bearing capacity (q_{ns})** : $q_{ns} = q_{nu} / F$ (F = factor of safety, usually 2.5 – 3)
- **Safe bearing capacity (q_s)** : $q_s = q_{ns} + \gamma D_f$
- **Gross safe bearing capacity** = q_u / F.
- **Safe / Allowable bearing pressure (q_a)** : the maximum net pressure which the soil can carry without shear failure AND without exceeding permissible settlement ⇒ q_a = smaller of (q_{ns}, pressure for tolerable settlement).

9. Terzaghi's Bearing Capacity Theory (1943)

Assumptions

- Footing is **strip (continuous)**, base is **rough**, soil is homogeneous, isotropic, **c–φ** soil.
- Shallow foundation : $D_f \leq B$; soil above base level replaced by uniform surcharge $q = \gamma D_f$ (its shear strength is neglected).
- Failure is by **general shear**; the Mohr–Coulomb criterion holds; load is vertical and concentric.

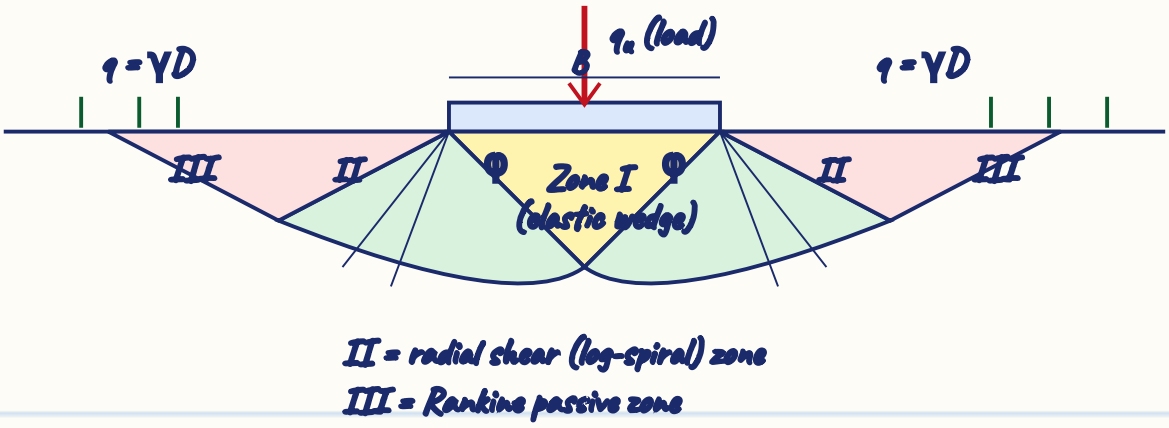


Fig. 2 – Terzaghi's failure mechanism (zones I, II, III) – base angles of wedge = φ

- **Zone I** : wedge below footing; moves down with footing; base angles = φ (rough base).
- **Zone II** : radial shear zone – curved boundary is a logarithmic spiral.
- **Zone III** : Rankine passive zones, base angles ($45° - \varphi/2$).

Outline of derivation (strip footing)

- Consider equilibrium of wedge I (per unit length). Forces: ① **downward** – $q_u \cdot B$ (load) + weight $W = \tfrac{1}{4}\gamma B^2 \tan\varphi$; ② **upward** – cohesion along two inclined faces = $2 \cdot C \cdot \sin\varphi$ where $C = c \cdot (B/2)/\cos\varphi$ per face ; and ③ passive reaction P_p on each face, inclined at φ to the normal.
- Vertical equilibrium $\Rightarrow q_u B + W - 2P_p - cB\tan\varphi = 0$
- P_p is made up of three parts: P_{pc} (due to cohesion c), P_{pq} (due to surcharge q) and $P_{p\gamma}$ (due to unit weight γ) $\Rightarrow$ solved with the log-spiral $\Rightarrow$ superposition gives :

$$q_u = c\,N_c + \gamma\,D_f\,N_q + \tfrac{1}{2}\gamma\,B\,N_\gamma$$

(strip footing, general shear failure)

- The 3 terms : cohesion component + surcharge component + self-weight (friction) component.
- N_c, N_q, N_γ = Terzaghi's bearing capacity factors – functions of φ only.

Shape Factors (Terzaghi)

Footing	q_u (general shear)
Strip	$cN_c + qN_q + 0.5\gamma BN_\gamma$
Square (B × B)	$1.3cN_c + qN_q + 0.4\gamma BN_\gamma$
Circular (dia B)	$1.3cN_c + qN_q + 0.3\gamma BN_\gamma$
Rectangular (B × L)	$cN_c(1+0.3B/L) + qN_q + 0.5\gamma BN_\gamma(1-0.2B/L)$

General vs. Local Shear Failure

- **General shear** (dense sand, stiff clay) : well-defined failure surface reaching ground; sudden failure, bulging, clear peak on the load–settlement curve.
- **Local shear** (loose sand, soft clay) : failure surface not continuous to the surface; gradual failure, no sharp peak – settlement is large.
- For local shear use reduced strength (Terzaghi) :

$$c' = (2/3)\,c \qquad \tan\varphi' = (2/3)\tan\varphi$$

- ...and use N'_c, N'_q, N'_γ (values at φ'). Eg. strip: $q'_u = (2/3)c\,N'_c + qN'_q + \tfrac{1}{2}\gamma BN'_\gamma$. Square: $q'_u = 0.867cN'_c + qN'_q + 0.4\gamma BN'_\gamma$.

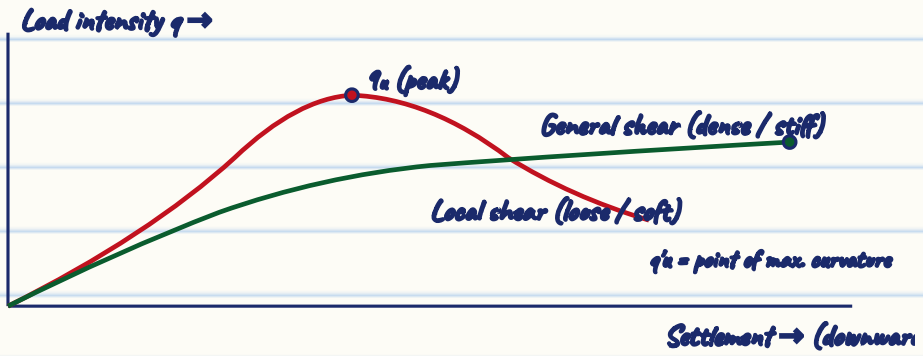


Fig. 3 – Load–settlement curves : general and local shear (axes drawn schematically)

Effect of Water Table (Terzaghi)

Submerged soil has $\gamma' = \gamma_{sat} - \gamma_w$ ($\approx \tfrac{1}{2}\gamma_{sat}$) → bearing capacity reduces. Let D_w = depth of W.T. below ground level.

- **Case 1 : W.T. above base level ($D_w \le D_f$)**
  - $q = \gamma D_w + \gamma'(D_f - D_w)$; use γ' in the 3rd term ; ($\gamma_w = 9.81\ kN/m^3$)
- **Case 2 : W.T. at base level ($D_w = D_f$)**
  - $q = \gamma D_f$ (no change) ; γ' in 3rd term.
- **Case 3 : W.T. below base but within B ($D_f < D_w < D_f + B$)**
  - $q = \gamma D_f$; γ in 3rd term replaced by average : $\boldsymbol{\gamma_{av} = \gamma' + (D_w - D_f)(\gamma - \gamma')/B}$
- **Case 4 : $D_w \ge D_f + B \Rightarrow$ no effect of W.T.**

Reduction-factor (IS 6403) method

$$q_u = cN_c + qN_q W' + \tfrac{1}{2}\gamma B N_\gamma W''$$

- $W' = 0.5\,[1 + D_{w1}/D_f]$ (W.T. depth D_{w1} measured from ground; $W' = 0.5$ if W.T. at ground level, 1 if at base or below).
- $W'' = 0.5\,[1 + D_{w2}/B]$ (D_{w2} = depth of W.T. below the base; $W'' = 0.5$ at base, 1 if $D_{w2} \ge B$).

Rise of water table to the base reduces q_u of a cohesionless soil by about 50 %. Always design for the worst (highest) W.T.

Terzaghi's Bearing Capacity Factors

$\varphi°$	0	5	10	15	20	25	30	35	40
N_c	5.7	7.3	9.6	12.9	17.7	25.1	37.2	57.8	95.7
N_q	1.0	1.6	2.7	4.4	7.4	12.7	22.5	41.4	81.3
N_γ	0	0.5	1.2	2.5	5.0	9.7	19.7	42.4	100.4

Worked Example 1 – Terzaghi (strip footing, c–φ soil)

Q. A strip footing 2 m wide is founded at 1.5 m depth in a soil with $c = 20\ kN/m^2$, $\varphi = 20°$, $\gamma = 18\ kN/m^3$. Find the safe bearing capacity for $F = 3$ (general shear). Take $N_c = 17.7$, $N_q = 7.4$, $N_\gamma = 5.0$.

Sol.

- $q = \gamma D_f = 18 \times 1.5 = 27\ kN/m^2$
- $q_u = cN_c + qN_q + \tfrac{1}{2}\gamma BN_\gamma$
- $= 20(17.7) + 27(7.4) + 0.5(18)(2)(5.0)$
- $= 354 + 199.8 + 90 = 643.8\ kN/m^2$
- $q_{nu} = q_u - \gamma D_f = 643.8 - 27 = 616.8\ kN/m^2$
- $q_{ns} = 616.8 / 3 = 205.6\ kN/m^2$

$$q_s = q_{ns} + \gamma D_f = 205.6 + 27 \approx 232.6\ kN/m^2$$

Worked Example 2 – Square footing with water table

Q. A square footing 2.5 m × 2.5 m is placed at 1.5 m depth in sand with $c = 0$, $\varphi = 30°$, $\gamma = 18\ kN/m^3$ above W.T. and $\gamma_{sat} = 20\ kN/m^3$. W.T. is at 1 m below ground level. Find the net safe bearing capacity ($F = 3$) and the safe load. ($N_q = 22.5$, $N_\gamma = 19.7$.)

Sol. W.T. above base ⇒ Case 1 ($D_w = 1\ m < D_f$).

- $\gamma' = 20 - 9.81 = 10.19\ kN/m^3$
- $q = \gamma D_w + \gamma'(D_f - D_w) = 18(1) + 10.19(0.5) = 23.1\ kN/m^2$
- Square footing, $c = 0$: $q_u = qN_q + 0.4\gamma' BN_\gamma$
- $= 23.1(22.5) + 0.4(10.19)(2.5)(19.7)$
- $= 519.6 + 200.7 = 720.4\ kN/m^2$
- $q_{nu} = 720.4 - 23.1 = 697.3\ kN/m^2 \Rightarrow q_{ns} = 697.3/3 = 232.4\ kN/m^2$

- Safe load = $q_{ns} \times B^2 = 232.4 \times 6.25 \approx$ **1452 kN** (net) [gross: $q_s = 255.5 \rightarrow$ 1597 kN]

Had the W.T. been at the base or deeper than B below it, qu would have been ≈ 40–50 % higher – it shows how important the water table is!

10. Meyerhof's Bearing Capacity Theory (1951, 1963)

- Improves Terzaghi : considers **shear strength of soil above the base** (failure surface extends to ground), shape, depth and load-inclination effects. Footing may be rough or smooth.
- Failure zones : elastic wedge + radial shear + mixed shear zone (a log-spiral surface rising to the ground surface – equivalent free surface).

General equation

$$q_u = c\,N_c\,s_c\,d_c\,i_c + q\,N_q\,s_q\,d_q\,i_q + \tfrac{1}{2}\gamma\,B\,N_\gamma\,s_\gamma\,d_\gamma\,i_\gamma$$

s = shape factors, d = depth factors, i = inclination factors. (For strip footing s = 1.) $q = \gamma D_f$

Bearing capacity factors (Meyerhof)

$$N_q = e^{\pi \tan\varphi}\tan^2(45^\circ + \varphi/2)\,;\; N_c = (N_q - 1)\cot\varphi\,;\; N_\gamma = (N_q - 1)\tan(1.4\varphi)$$

φ°	0	5	10	15	20	25	30	35	40
N_c	5.14	6.49	8.35	10.98	14.83	20.72	30.14	46.12	75.31
N_q	1.0	1.57	2.47	3.94	6.40	10.66	18.40	33.30	64.20
N_γ	0	0.45	1.22	2.65	5.39	10.88	22.40	48.03	109.4

Shape, depth & inclination factors

Let $K_p = \tan^2(45^\circ + \varphi/2)$ (coefficient of passive pressure), α = inclination of load with the vertical.

Factor	$\varphi = 0$	$\varphi > 10°$
Shape s_c	$1 + 0.2\,(B/L)\,K_p$	
$s_q = s_\gamma$	1	$1 + 0.1\,(B/L)\,K_p$
Depth d_c	$1 + 0.2\,(D_f/B)\,\sqrt{K_p}$	
$d_q = d_\gamma$	1	$1 + 0.1\,(D_f/B)\,\sqrt{K_p}$
Inclination $i_c = i_q$	$(1 - \alpha°/90°)^2$	
i_γ	0 (for $\varphi = 0$)	$(1 - \alpha°/\varphi°)^2$

For $\varphi = 0$ clay : $s_c = 1 + 0.2B/L$. For strip ($B/L = 0$) all shape factors = 1; for vertical load $i = 1$.

II. Skempton's Method (for Clays, $\varphi_u = 0$)

- For **saturated clays under undrained loading**; $c = c_u$, $\varphi = 0 \Rightarrow N_q = 1$, $N_\gamma = 0$.
- Skempton (1951) showed that N_c is not constant – it increases with the depth/width ratio D_f/B (because of the increase of shear resistance of the soil above base level) and with footing shape.

$$q_{nu} = c_u N_c \quad \Rightarrow \quad q_u = c_u N_c + \gamma D_f$$

Skempton's N_c values

- **Strip** : $N_c = 5\,(1 + 0.2\,D_f/B)$ for $D_f/B \leq 2.5$; $N_c = 7.5$ for $D_f/B > 2.5$
- **Square / circular** : $N_c = 6\,(1 + 0.2\,D_f/B)$ for $D_f/B \leq 2.5$; $N_c = 9$ for $D_f/B > 2.5$
- **Rectangular** ($B \times L$) : N_c (rect) = N_c (strip) · $(0.84 + 0.16\,B/L)$

D_f/B	0	0.5	1.0	1.5	2.0	2.5	≥ 4
N_c strip	5.0	5.5	6.0	6.5	7.0	7.5	7.5
N_c square	6.0	6.6	7.2	7.8	8.4	9.0	9.0

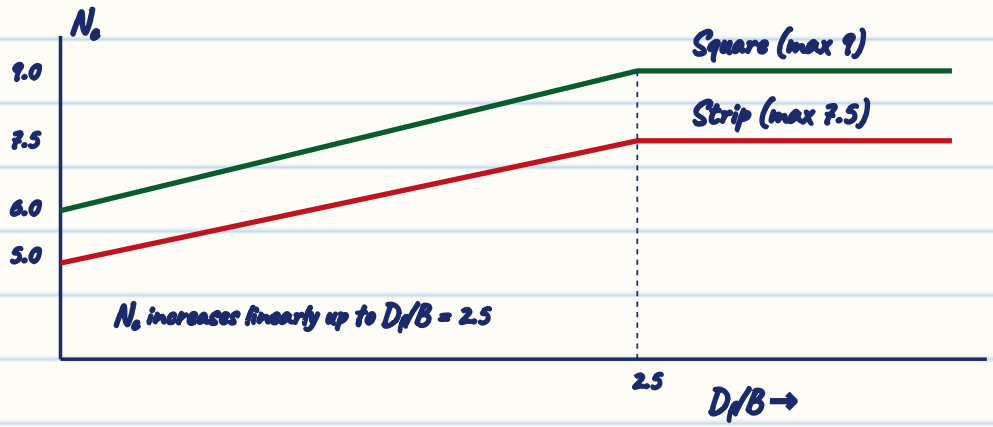


Fig. 4 – Skempton's N_c vs D_f/B (schematic)

Net safe bearing capacity

$$q_{ns} = c_u N_c / F \;;\; q_s = q_{ns} + \gamma D_f$$

- F = 2.5 to 3. For a surface footing on clay (strip) : $q_{nu} = 5.14\, c_u$ (Prandtl / Meyerhof) ≈ $5\, c_u$ (Skempton).
- Eg: c_u = 60 kN/m², square footing B = 2 m, D_f = 1 m, γ = 18 : D_f/B = 0.5 ⇒ N_c = 6.6; q_{nu} = 396; q_{ns} = 132; q_s = 132 + 18 = 150 kN/m².

ALLOWABLE BEARING PRESSURE

12. Safe Bearing Pressure from N-value (SPT)

- SPT (IS 2131) gives N = no. of blows for 300 mm penetration of a split spoon – a quick, in-situ indicator of density (sands) & consistency (clays).
- In sands, settlement (not shear) usually governs ⇒ allowable pressure is based on **25 mm** (or 40 mm) of settlement.

Corrections to field N (IS 2131)

- **Overburden** : $N_c = C_N N$, $C_N = 0.77 \log_{10}(2000/\sigma'_v)$ (σ'_v in kN/m², for $\sigma'_v \geq 25$)
- **Dilatancy** (saturated fine / silty sand, $N > 15$) : $N_c = 15 + ½ (N - 15)$
- Also : energy, rod length, borehole & sampler corrections in N_{60}.

Sands			Clays		
N	Density	φ°	N	Consistency	q_u kN/m²
0–4	V. loose	< 29	< 2	V. soft	< 25
4–10	Loose	29–30	2–4	Soft	25–50
10–30	Medium	30–36	4–8	Medium	50–100
30–50	Dense	36–41	8–15	Stiff	100–200
> 50	V. dense	> 41	15–30	V. stiff	200–400

Peck's approximation : $\varphi \approx 27 + 0.3\,N$ (degrees) ; for clays $c_u = q_u/2 \approx 6N$ to $7N$ (kN/m²)

(i) Terzaghi–Peck method

- Based on charts of allowable pressure vs. footing width B for different N (for 25 mm settlement). Footing on sand : q_a increases with B for shear but decreases for settlement ⇒ for B > ~1.2 m, q_a becomes nearly constant.
- Water table correction : if W.T. is at / above base, multiply q_a by **0.5** (W.T. at base ≈ 0.5 – 0.6 effect).

(ii) Meyerhof's equations (25 mm settlement, N_{60})

$$B \le 1.2\ \text{m}: \quad q_a\,(\text{net}) = 12\,N\,K_d \ \text{kN/m}^2$$
$$B > 1.2\ \text{m}: \quad q_a\,(\text{net}) = 8\,N\,[(B + 0.3)/B]^2\,K_d \ \text{kN/m}^2$$

- $K_d = 1 + 0.33\,D_f/B \le 1.33$ (depth factor). N = average SPT value over depth B below base (corrected). Raise by 50 % for 40 mm settlement. Halve the value if W.T. is at the base level.

13. Plate Load Test (IS 1888 – 1982)

- A **field test** to obtain ultimate bearing capacity and settlement of soil at the foundation level by loading a rigid plate (model footing). Test is done in a pit of size ≥ 5 × plate width at the proposed level.
- Plate : mild steel, 25 mm thick, **square 300–750 mm** or circular of 300 – 750 mm dia. (usually 300 mm for sand; 450 – 600 for clay).

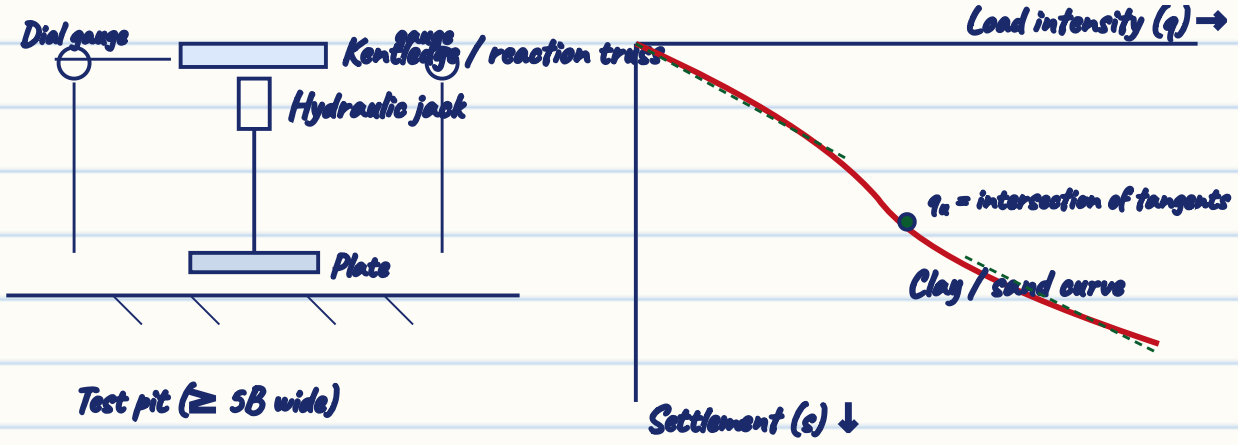


Fig. 5 – Plate load test arrangement and load–settlement curve

Procedure

1. Excavate pit, level the bottom; place plate on thin sand bed, seat it with a seating load of 7 kN/m² and release.
2. Apply load in increments of ≈ 1/5 of the estimated safe load (or 1/10 of ultimate), through a hydraulic jack reacting against kentledge / anchored truss.
3. Record settlement by 2–4 dial gauges (0.01 mm) at time intervals 1, 2.25, 4, 6.25, 9, 16, 25 … min; next load applied when rate of settlement < 0.02 mm/min (or after 1 hr).
4. Continue till failure / total settlement of 25 mm (or 1/10 B) or twice the design intensity.

Interpretation

- Plot load intensity q vs. settlement s (arithmetic and log–log). q_u is at the break point (intersection of tangents). If no break : q at s = 25 mm / 10 % of B_p.
- **Bearing capacity (size effect):** Clay : $q_f = q_p$ (independent of size) ; Sand : $q_f = q_p \cdot (B_f/B_p)$
- Safe B.C. = q_u/F ; and settlement governs.

Settlement of footing from plate settlement ($S_p \rightarrow S_f$)

$$\text{Sand}: \quad S_f = S_p \cdot [\, B_f (B_p + 0.3) \,/\, (\, B_p (B_f + 0.3)\,)\,]^2$$

$$\text{Clay}: \quad S_f = S_p \cdot (B_f / B_p)$$

(B in metres; same intensity of loading q for plate and footing.)

Example 4. A 300 mm square plate settles 12 mm under a certain pressure on sand. Find the settlement of a 2 m square footing at the same pressure.

- $S_f = 12 \times [2\,(0.3 + 0.3) / (0.3\,(2 + 0.3))]^2 = 12 \times (1.2 / 0.69)^2 = 12 \times 3.025 \approx$ 36.3 mm

Limitations of plate load test

- Plate stress bulb $\approx 2B_p$ deep – does **not** test deeper layers (weak layer below remains undetected).
- Does not give long-term consolidation settlement in clays (test lasts hours).
- Scale effect; ground-water level effect; costly & time-consuming.

14. Allowable (Permissible) Settlements

- **Total settlement (S_t)** = immediate (S_i) + primary consolidation (S_c) + secondary compression (S_s).
- **Differential settlement (δ)** : difference in settlement between two points of the same structure; **angular distortion (β)** = δ / L (L = distance between the points).
- Differential settlement – causes cracks, tilting, distress; more serious than uniform settlement. Uniform settlement (even 100 mm) is usually harmless except for connecting services.
- Limits depend on structure type, rigidity, soil, and use. Sands : $S_{diff} \approx 0.5$–$0.75\ S_{max}$; clays : $S_{diff} \approx 0.5\ S_{max}$ (Skempton & MacDonald).

IS 1904 – 1986 permissible settlements

Foundation	Soil	Max. S (mm)	Differential	Ang. dist.
Isolated footing	Sand / hard clay	50	0.0033 L	1/300
	Plastic clay	75	0.0033 L	1/300
Raft	Sand / hard clay	75	0.0033 L	1/300
	Plastic clay	100	0.0033 L	1/300

Angular distortion limits (Bjerrum, 1963)

$\beta = \delta/L$	Effect
1/750	Limit for machinery sensitive to settlement
1/500	Safe limit for framed buildings with no cracking
1/300	Cracking of panel / partition walls begins
1/250	Tilting of tall rigid buildings visible
1/150	Structural damage to general buildings